DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)

Project Synopsis

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Project Title: Financial Transaction Fraud Detection and Risk Intelligence Platform

Introduction: The rapid growth of digital banking, online payments, e-commerce, and electronic financial transactions has increased the convenience of financial services while also creating opportunities for fraudulent activities. Financial fraud can result in significant monetary losses, compromised customer accounts, and reduced trust in digital payment systems. Traditional fraud detection mechanisms often rely on predefined rules and threshold-based approaches, which may be ineffective against evolving and complex fraudulent transaction patterns.

Machine Learning (ML) techniques provide the ability to analyze large volumes of historical transaction data and identify patterns associated with fraudulent behaviour. However, financial transaction datasets are often highly imbalanced, with legitimate transactions greatly outnumbering fraudulent transactions. This makes accurate fraud detection challenging and requires appropriate preprocessing, feature engineering, model selection, and evaluation.

The proposed project develops a Financial Transaction Fraud Detection and Risk Intelligence Platform that analyzes transaction characteristics and transaction patterns to identify potentially fraudulent transactions. The system applies machine learning models for fraud classification and generates a risk score to indicate the likelihood of a transaction being suspicious. A centralized database and monitoring dashboard are used to maintain transaction records, visualize risk trends, and generate alerts for high-risk transactions.

Objectives:

- To design and develop a financial transaction monitoring and fraud detection platform.
- To preprocess and analyze financial transaction data for identifying relevant fraud-related patterns.

- To develop and compare suitable machine learning models for fraudulent transaction detection.
- To address class imbalance using appropriate data preprocessing and resampling techniques.
- To generate a risk score based on transaction characteristics and fraud prediction results.
- To provide alerts and visual analytics for suspicious and high-risk transactions.
- To maintain transaction and fraud-related records for historical analysis and monitoring.
- To evaluate the proposed system using appropriate performance metrics such as precision, recall, F1-score and ROC-AUC.

Software Requirements

Operating System

- Windows 11 / Ubuntu Linux / macOS

Programming Language

- Python 3.11+
- JavaScript

Machine Learning & Data Processing

- Scikit-learn
- NumPy
- Pandas
- Imbalanced-learn

Explainability and Visualization

- SHAP
- Matplotlib
- Seaborn

Backend

- Flask / FastAPI
- REST APIs

Frontend

- React.js

Database

- MySQL / PostgreSQL

Development Tools

- Visual Studio Code
- Jupyter Notebook / Google Colab
- Git and GitHub

Deployment

- Local deployment

Hardware Requirements

- Computer / Laptop with minimum 8 GB RAM
- Minimum Intel Core i5 / AMD Ryzen 5 or equivalent processor
- Minimum 256 GB available storage

Connectivity & Deployment

- Stable Internet connectivity

- Optional cloud infrastructure for deployment

Abstract: Financial fraud has become an important concern with the rapid adoption of digital banking, online shopping, electronic payments, and card-based transactions. Traditional rule-based fraud detection systems may have difficulty identifying evolving fraud patterns and can generate a large number of false alerts.

This project proposes a Financial Transaction Fraud Detection and Risk Intelligence Platform using Machine Learning. The system processes transaction data and performs preprocessing, feature engineering, and fraud classification using suitable machine learning models. Since fraudulent transactions usually represent a very small proportion of the overall transaction data, class imbalance handling techniques are incorporated during model development and evaluation.

In addition to binary fraud classification, the proposed system generates a risk score to categorize transactions according to their potential risk. The platform provides a monitoring dashboard for viewing transaction statistics, suspicious activities, fraud trends, and high-risk transactions. Alerts can be generated when transactions exceed predefined risk thresholds. Transaction and prediction records are maintained in a database to support historical analysis and monitoring.

The system aims to provide a modular and extensible platform for financial fraud analysis that combines machine learning-based detection, risk assessment, data visualization, and transaction monitoring in a unified system.

Literature Survey:

Sl. No	Paper Title	Authors , Year & Publisher	Methodology	Limitation
1	**Ensemble Learning based Credit Card Fraud Detection System**	**Authors:** Pooja Tomar, Sonika Shrivastava, Urjita Thakar **Year:** 2021 **Publisher:** IEEE, 5th Conference on Information and Communication Technology (CICT)	The study applies an ensemble learning approach using Decision Tree, Logistic Regression and Naive Bayes classifiers with hard voting to improve fraud detection.	The work primarily focuses on classification performance and does not provide a complete transaction risk intelligence platform incorporating risk scoring, monitoring and alert management.
2	**Customer Transaction Fraud Detection Using XGBoost Model**	**Authors:** Yixuan Zhang, Jialiang Tong, Ziyi Wang, Fengqiang Gao **Year:** 2020 **Publisher:** IEEE, International Conference on Computer Engineering and Application (ICCEA)	The study applies the XGBoost machine learning algorithm for identifying fraudulent customer transactions.	The work focuses primarily on transaction fraud classification using XGBoost rather than providing an integrated risk scoring and transaction monitoring platform.

3	**Fraud Detection in Credit Card Data using Unsupervised Machine Learning Based Scheme**	**Authors:** Arun Kumar Rai, Rajendra Kumar Dwivedi **Year:** 2020 **Publisher:** IEEE, International Conference on Electronics and Sustainable Communication Systems (ICESC)	The study investigates an unsupervised machine learning approach for identifying fraudulent patterns in credit card transaction data.	The approach is primarily focused on fraud detection from transaction data and does not provide an integrated application-level risk monitoring and alerting system.
4	**Interpretable Credit Card Fraud Detection Using Machine Learning Leveraging SHAP**	**Authors:** Joy Biswas, Abir Ahmed Mridha, Mohammad Sakib Hossain, Ananya Subhra Trisha, Md. Sabbir Ahmed, Md. Iqbal Hossain **Year:** 2023 **Publisher:** IEEE, 6th International Conference on Electronic Information and Communication Technology (ICEICT)	The study evaluates Logistic Regression, Random Forest, AdaBoost and hybrid models and uses SHAP to explain model predictions.	The work primarily focuses on explainable fraud classification and does not cover a complete transaction monitoring platform with integrated risk scoring, dashboards and alert management.
5	**Advancing Credit Card Fraud Detection Through Explainable Machine Learning Methods**	**Authors:** Vikas Adhegaonkar, Abhijeet Thakur, Nikhil Varghese **Year:** 2024 **Publisher:** IEEE, 2nd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)	The study evaluates Decision Tree, Logistic Regression, Support Vector Machine and Random Forest models and incorporates explainable machine learning for fraud analysis.	The research focuses on model-based fraud classification and explainability rather than implementing a broader transaction risk intelligence and monitoring platform.
6	**Transparency and Privacy: The Role of Explainable AI and Federated Learning in Financial Fraud Detection**	**Authors:** Tomisin Awosika, Raj Shukla, Bernardi Pranggono **Year:** 2024 **Publisher:** IEEE Access	The study investigates explainable AI and federated learning for financial fraud detection, with particular emphasis on data imbalance, transparency and privacy.	Federated learning introduces additional architectural complexity and is primarily investigated from the privacy-preserving and distributed-learning perspective rather than as an end-to-end transaction risk monitoring platform.

Existing System & its Drawback:

Existing System: Existing financial fraud detection systems commonly use rule-based mechanisms, statistical analysis, and machine learning models to identify suspicious transactions. Rule-based systems typically rely on predefined conditions such as transaction amount thresholds, unusual locations, transaction frequency, or other manually defined indicators. Machine learning-based systems analyze historical transaction data to classify transactions as legitimate or fraudulent.

Several existing research systems focus primarily on improving the accuracy of fraud classification using individual machine learning models, ensemble methods, or specialized techniques for handling class imbalance.

Drawbacks:

- Dependence on predefined rules may make traditional systems difficult to adapt to evolving fraud patterns.
- Highly imbalanced transaction datasets make accurate identification of fraudulent transactions challenging.
- High false-positive rates may result in legitimate transactions being incorrectly flagged.
- Binary fraud classification alone does not indicate the degree of risk associated with a transaction.
- Many research implementations focus primarily on model performance rather than complete transaction monitoring.
- Limited integration of fraud prediction with risk scoring, alerts, visualization, and historical transaction analysis.
- Model predictions may be difficult to interpret without appropriate explainability mechanisms.
- Fraud patterns can change over time, requiring continuous monitoring and model evaluation.

Proposed System & Advantages:

Proposed System: The proposed Financial Transaction Fraud Detection and Risk Intelligence Platform integrates transaction processing, machine learning-based fraud detection, risk scoring, database management, and visualization into a unified system.

Transaction data is first collected and preprocessed to handle missing values, normalization, feature transformation, and class imbalance. Relevant transaction features are then provided to selected machine learning models for fraud classification. The prediction output is combined with transaction risk indicators to generate a risk score representing the likelihood that a transaction is suspicious.Wherever applicable, explainability techniques such as SHAP may be used to identify the contribution of relevant features to individual model predictions.

The system stores transaction and prediction records in a centralized database and provides a dashboard for monitoring transaction statistics, fraud trends, risk levels, and suspicious activities. High-risk transactions can trigger alerts for further investigation. The modular architecture allows different machine learning models and risk-assessment techniques to be evaluated and incorporated with minimal changes to the overall system.

Advantages:

- Automated detection of potentially fraudulent transactions.
- Risk scoring provides more information than simple fraud/not-fraud classification.
- Supports analysis of highly imbalanced financial transaction data.
- Combines fraud detection with transaction monitoring and visualization.
- Provides alerts for high-risk transactions.
- Maintains historical transaction and prediction records.
- Allows comparison and evaluation of multiple machine learning models.
- Modular architecture allows future enhancement of detection techniques.
- Provides a practical demonstration of ML integrated with backend, database and frontend technologies.

SDG Mapping:

SDG No. 8: Decent Work and Economic Growth

Goal: Promote sustained, inclusive and sustainable economic growth, productive employment and decent work for all.

Relevance: The proposed system contributes to safer digital financial transactions by identifying potentially fraudulent activities and reducing financial risks. Improved fraud monitoring can help strengthen trust in digital financial services and support secure participation in the digital economy.

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